

 psundaravadivel / Ex.-No-9---Packet-Tracer-Subnet-an-IPv4-Network

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Ex. No: 9 - Packet Tracer: Subnet an IPv4 Network

Date: 16-09-2026

Objective

Design, configure, and verify an IPv4 subnetting scheme in Cisco Packet Tracer.

- Subnet the 192.168.0.0/24 network into multiple subnets.
- Assign addresses to LAN-A, LAN-B, and future networks.
- Configure IP addressing on routers, switches, and PCs.
- Verify connectivity using ping and router show commands.

Apparatus / Tools Required

- Cisco Packet Tracer
 - CustomerRouter (2911 or equivalent)
 - ISP Router
 - 2 Customer Switches (LAN-A Switch, LAN-B Switch)
 - ISP Switch
 - 2 PCs (PC-A, PC-B)
 - ISP Workstation and ISP Server
 - Copper straight-through cables for LAN links
 - Serial DCE/DTE cable for WAN link
-

Network Topology Diagram

(Insert your Packet Tracer screenshot showing CustomerRouter → LAN-A Switch → PC-A, CustomerRouter → LAN-B Switch → PC-B, and ISP side with Router, Switch, Workstation, Server, and Serial link.)

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
CustomerRouter	G0/0	(1st host of LAN-A subnet)	(Subnet mask)	N/A
CustomerRouter	G0/1	(1st host of LAN-B subnet)	(Subnet mask)	N/A
CustomerRouter	S0/1/0	209.165.201.2	255.255.255.252	N/A
LAN-A Switch	VLAN1	(2nd host of LAN-A subnet)	(Subnet mask)	CustomerRouter G0/0
LAN-B Switch	VLAN1	(2nd host of LAN-B subnet)	(Subnet mask)	CustomerRouter G0/1
PC-A	NIC	(Last host of LAN-A subnet)	(Subnet mask)	CustomerRouter G0/0
PC-B	NIC	(Last host of LAN-B subnet)	(Subnet mask)	CustomerRouter G0/1
ISP Router	G0/0	209.165.200.225	255.255.255.224	N/A
ISP Router	S0/1/0	209.165.201.1	255.255.255.252	N/A
ISP Switch	VLAN1	209.165.200.226	255.255.255.224	209.165.200.225
ISP Workstation	NIC	209.165.200.235	255.255.255.224	209.165.200.225
ISP Server	NIC	209.165.200.240	255.255.255.224	209.165.200.225

(LAN-A and LAN-B IPs to be filled in after subnetting calculation.)

Procedure

Part 1: Subnet the Assigned Network

1. Start with 192.168.0.0/24.
2. Requirements:
 - o LAN-A: ≥ 50 hosts
 - o LAN-B: ≥ 40 hosts
 - o At least 2 unused subnets for future expansion.
3. Choose a subnet mask that supports both host requirements.
 - o Example: /26 (255.255.255.192) $\rightarrow$ 62 hosts per subnet, 4 subnets.

4. Allocate:

- o Subnet 1 → LAN-A
 - o Subnet 2 → LAN-B
 - o Subnets 3 & 4 → Reserved
-

Part 2: Configure the Devices

CustomerRouter

- Set hostname: hostname CustomerRouter
 - Passwords:
 - o enable secret Class123
 - o line console 0 → password Cisco123 → login
 - Configure interfaces:
 - interface g0/0
 - ip address
 - no shutdown
 - interface g0/1
 - ip address
 - no shutdown
 - interface s0/1/0
 - ip address 209.165.201.2 255.255.255.252
 - clock rate 64000 ← DCE end
 - no shutdown
 - Save: copy running-config startup-config
 - Switches (LAN-A, LAN-B)
 - VLAN1 IP: 2nd host of subnet
 - Default gateway: Router G0/0 or G0/1
 - PCs (PC-A, PC-B)
 - Last host of subnet
 - Subnet mask and gateway configured
-

Part 3: Verification & Testing

1. On routers:

- o show ip interface brief (check status up/up)
- o show ip route (verify connected subnets)

2. On PCs:

- o Ping default gateway
 - o Ping across subnets (PC-A ↔ PC-B)
 - o Ping ISP Server
-

Commands Used (Summary)

- Mode/navigation: enable, configure terminal, end
 - Interface config: interface g0/0, ip address, no shutdown
 - Show/verify: show ip interface brief, show ip route
 - Save: copy running-config startup-config
-

Output (Attach Screenshots)

Cisco Packet Tracer - C:/Users/acer/Downloads/11.5.5-packet-tracer---subnet-an-ipv4-network.pka - niha! mohamed B - 2026-08-25 10:33:37

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1318, y: 42

Root 05:29:03

```
graph LR
    PC_A[PC-A 50 Hosts] --- LAN_A[LAN-A]
    LAN_A --- CustomerRouter
    PC_B[PC-B 40 Hosts] --- LAN_B[LAN-B]
    LAN_B --- CustomerRouter
    CustomerRouter --- ISPRouter
    ISPRouter --- ISP_Switch[ISP Switch]
    ISP_Switch --- ISP_Workstation[ISP Workstation]
    ISP_Switch --- ISP_Server[ISP Server]
```

PT Activity: 00:59:22

Packet Tracer – Subnet an IPv4 Network

Addressing Table

Device	Interface	IP Address	Subnet Ma
CustomerRouter	G0/0		
	G0/1		
	S0/1/0	209.165.201.2	255.255.255.2
LAN-A Switch	VLAN1		
LAN-B Switch	VLAN1		
PC-A	NIC		
PC-B	NIC		

Time Elapsed: 00:59:22 Completion: 100%

☐ Top ☐ Dock 1/1

Time: 00:58:39

PT8200 4331 4321 IR8340 1941 2901 2911 8191OX 8191GW 829 1240

Search for device

2901

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
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Realtime Simulation

PC	NIC	IP	Subnet	Gateway
		10.1.2.10	255.255.255.0	10.1.2.1

Objectives

- Part 1: Display Router Information
- Part 2: Configure Router Interfaces
- Part 3: Verify the Configuration

Background

In this activity, you will use various **show** commands to display the current state of the router. You will then use the Addressing Table to configure router Ethernet interfaces. Finally, you will use commands to verify and test your configurations.

Note: The routers in this activity are partially configured. Some of the configurations are not covered in this course but they are provided to assist you in using verification commands.

Part 1: Display Router Information

Step 1: Display interface information on R1.

Note: Click a device and then click the CLI tab to access the command line directly. The console password is **cisco**. The privileged EXEC password is **class**.

- Which command displays the statistics for all interfaces configured on a router?
- Which command displays the information about the Serial 0/0/0 interface only?

Time Elapsed: 00:51:10 Completion: 100%

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Result

The IPv4 subnetting scheme was successfully designed and implemented. Router, switches, and PCs were configured with correct addressing. Connectivity within LANs, across subnets, and to ISP devices was verified using ping and show commands.